

Ex. $V = \{y \mid y'' + 2y' = 0\}$

$W = \{f \mid f''' + f' = 0\}$

Find $\text{Hom}_{\mathbb{R}}(V, W) = \{\sigma: V \rightarrow W \mid \sigma(a_1 v_1 + a_2 v_2) = a_1 \sigma(v_1) + a_2 \sigma(v_2)\}$

$V = \{c_1 e^{-2x} + c_2 \mid c_1, c_2 \in \mathbb{R}\}$

$W = \{d_1 \cos x + d_2 \sin x \mid d_1, d_2 \in \mathbb{R}\}$

$\sigma: V \longrightarrow W$

$c_1 e^{-2x} + c_2 \longmapsto d_1 \cos x + d_2 \sin x$

$d_1 = \rho_1(c_1, c_2)$

$d_2 = \rho_2(c_1, c_2)$

$\sigma(c_1 e^{-2x} + c_2) = \sigma(c_1 e^{-2x}) + \sigma(c_2 \cdot 1)$

$= c_1 \sigma(e^{-2x}) + c_2 \sigma(1)$

$= c_1 (d_{11} \cos x + d_{21} \sin x) + c_2 (d_{12} \cos x + d_{22} \sin x)$

$= (\cos x \sin x) \begin{pmatrix} c_1 d_{11} + c_2 d_{12} \\ c_1 d_{21} + c_2 d_{22} \end{pmatrix}$

$= (\cos x \sin x) \begin{pmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$

$(e^{-2x} \ 1) \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} \longmapsto (\cos x \sin x) A_{2 \times 2} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix}$

Recall CH III Linear Maps

Isomorphism $\sigma \in \text{Hom}(V, W) \iff \dim V = \dim W$ (suppose $\dim V, \dim W < \infty$)

$\iff V \cong \mathbb{R}^n \cong W$

Want to know $\forall \sigma \in \text{Hom}(V, W)$

Fix basis $v_1, v_2, \dots, v_n$ of V

$w_1, w_2, \dots, w_m$ of W

$\forall v \in V$ want $\sigma(v)$

suppose $v = x_1 v_1 + x_2 v_2 + \dots + x_n v_n = (v_1, v_2, \dots, v_n) \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$

Then $\sigma(v) = \sigma\left(\sum_{i=1}^n x_i v_i\right) = \sum_{i=1}^n x_i \sigma(v_i)$

suppose $\sigma(v_1) = p_{11} w_1 + p_{21} w_2 + \dots + p_{m1} w_m$

$\vdots$
 $\sigma(v_n) = p_{1n} w_1 + p_{2n} w_2 + \dots + p_{mn} w_m$

Written as $(\sigma(v_1), \sigma(v_2), \dots, \sigma(v_n)) = (w_1, w_2, \dots, w_m)$ $\begin{pmatrix} P_{11} & \dots & P_{1n} \\ P_{21} & \dots & P_{2n} \\ \vdots & & \vdots \\ P_{m1} & \dots & P_{mn} \end{pmatrix}$

Then $\sigma(v) = (\sigma(v_1), \sigma(v_2), \dots, \sigma(v_n)) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ P_σ transition matrix
过渡矩阵

$$= [w_1 \dots w_m] P_{m \times n} X$$

$$= [w_1, w_2, \dots, w_m] \begin{pmatrix} \text{基} & \text{坐标} \\ P_{m \times n} X \end{pmatrix}$$

• TM (FTOLA, Fundamental Theorem of LA)

Let $\dim V = n$, $\dim W = m$. Then $(\text{Hom}(V, W), +, \cdot) \cong (\mathbb{F}^{m \times n}, +, \cdot)$

• Establish "+", "·" over $\text{Hom}(V, W) \rightarrow \sigma, \tau$

$$"+": \sigma + \tau \quad V \longrightarrow W$$

$$v \longmapsto (\sigma + \tau)(v) = \sigma(v) + \tau(v)$$

$$"\cdot": \forall a \in \mathbb{F}, a\sigma: v \longrightarrow (a\sigma)(v) = a\sigma(v)$$

Easy to see that $\sigma + \tau$ and $a\sigma \in \text{Hom}(V, W)$

Further more, $(\text{Hom}(V, W), +, \cdot)$ is a v.s. / $\mathbb{F}$

All + 0 are clear since $0: v \longrightarrow 0$ (zero map)

$$\text{Princess is } (-\sigma): v \longrightarrow -(\sigma(v))$$

$$\text{RK1 } (-\sigma)(v) = -(\sigma(v))$$

RK2 $\text{Hom}(V, W)$ is a v.s. / $\mathbb{F}$ for all V, W

$$\text{pf: } (\text{Hom}(V, W), +, \cdot) \cong (\mathbb{F}^{m \times n}, +, \cdot) = \text{Hom}(\mathbb{F}^n, \mathbb{F}^m)$$

$$\text{Define } \phi: \text{Hom}(V, W) \longrightarrow \mathbb{F}^{m \times n}$$

$$\sigma \longmapsto P_\sigma$$

$$\text{step 1 } \phi \in \text{Hom}(\text{Hom}(V, W), \mathbb{F}^{m \times n})$$

$$\phi(\sigma + \tau) = \phi(\sigma) + \phi(\tau)$$

$$P_{\sigma + \tau} = P_\sigma + P_\tau$$

$$P_{\sigma + \tau} e_i = \text{coordinate of } (\sigma + \tau)(v_i)$$

$$= \text{coordinate of } \sigma(v_i) + \text{coordinate of } \tau(v_i)$$

$$= P_\sigma e_i + P_\tau e_i$$

$$P_{a\sigma} = a P_\sigma$$

Step 2. ϕ is injective

Let $\sigma \in \ker \phi$ i.e. $\phi(\sigma) = 0 \Rightarrow \sigma = 0$

ϕ is surjective

$\forall A \in \mathbb{F}^{m \times n}$, find $\sigma \in \text{Hom}(V, W)$ s.t. $P\sigma = A$

Define $\sigma(v) = (w_1, \dots, w_m)Ax$